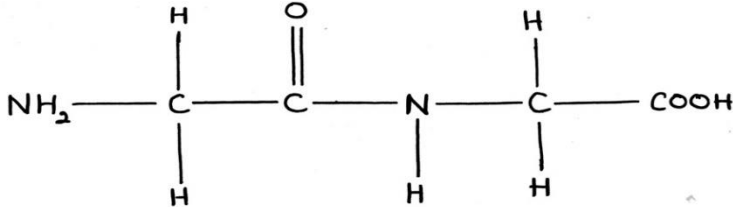


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
11.	(a)	(i)				1				1
		(ii)		formed from two molecules of the same acid	1					1
	(b)	(i)	I	$^+\text{NH}_3 - \text{CH}_2 - \text{COO}^-$	1					1
			II	it does not have an acidic hydrogen atom (in the 'carboxylic acid' group)			1			1
		(ii)	I	$\text{C}_3\text{H}_5\text{NO}^+ / \text{CH}_2\text{-C(O)-N(H)-CH}_2^+$			1			1
			II	award (1) for two correct absorption values / ranges N—H ~3000-3500 cm^{-1} C=O ~ 1650-1750 cm^{-1}		1				1
	(c)	(i)		0.53		1			1	1
		(ii)		dye absorbs in the yellow / green region colours seen comprises the rest / is the complimentary colour / is made from red and blue			1			2